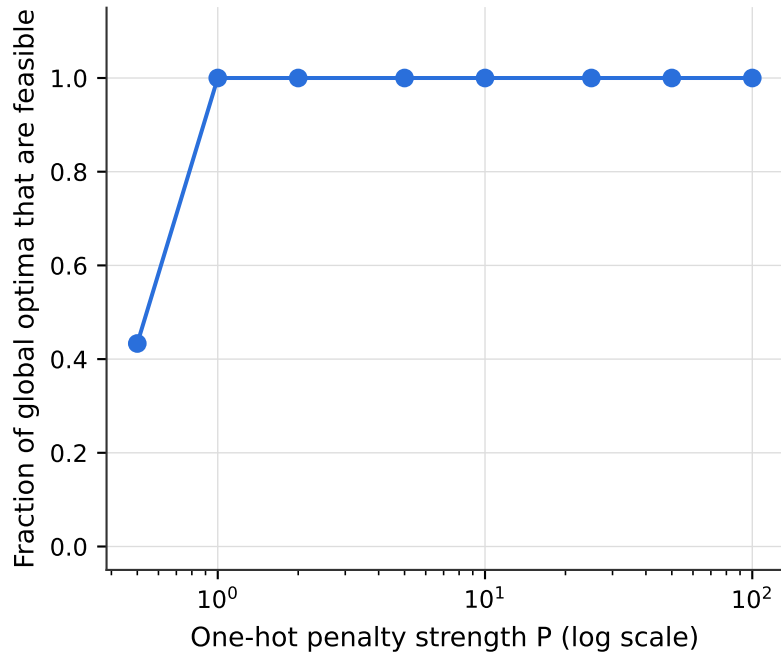


Penalty-sensitivity sweep: global QUBO minimum stays feasible across P

(a) One-hot penalty sensitivity



(b) Security-floor penalty sensitivity

